

A Standalone LoRa Mesh Network with LoRa-Optimized AODV and End-to-End Encrypted Payloads on MCU-Class Devices

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Abstract—Low-power long-range (LoRa) radios are attractive for communication in disaster zones, rural regions, and infrastructure-poor environments, but LoRa's extremely low bitrate, lack of native routing, and limited built-in end-to-end protection for standalone multi-hop operation limit its ability to form reliable mesh networks. This paper presents RelayCom, a fully standalone LoRa mesh system running on ultra-constrained microcontrollers (Arduino Nano + RYLR890) and providing end-to-end payload confidentiality, while embedding a LoRa-aware AODV routing optimization that significantly reduces control overhead. Unlike classical AODV, RelayCom introduces a unicast-first RREQ strategy, falling back to broadcast only after a timeout, preserving LoRa bandwidth and reducing collision probability. RelayCom integrates AES-CTR encryption at the firmware level to provide payload confidentiality across multi-hop forwarding. A complete modular firmware (LoRa handler, routing manager, state machine, and message buffer) is implemented and evaluated across multiple routing scenarios. Experimental results using a real three-node testbed demonstrate successful multi-hop route discovery, encrypted message forwarding, reduced route-discovery broadcast transmissions, and robust operation under constrained radio conditions. RelayCom illustrates a practical and low-cost blueprint for resilient, confidentiality-preserving, and infrastructure-independent LoRa mesh networks suitable for emergency communication and off-grid deployment.

Index Terms—LoRa mesh; AODV; route discovery; broadcast reduction; AES-CTR; end-to-end encryption; Arduino Nano; low-power embedded networking.

I. INTRODUCTION

Low-power wide-area (LPWA) technologies such as LoRa have emerged as a compelling solution for long-range, energy-efficient wireless communication in scenarios where traditional infrastructure is unavailable or unreliable. Their robustness and large coverage radius make them particularly suitable for emergency response, remote sensing, and off-grid communication systems. However, conventional deployments—typically based on star-topology LoRaWAN—depend on centralized gateways, which limits applicability in infrastructure-less environments [1].

When nodes must self-organize without gateways, multi-hop mesh routing becomes essential. Yet, LoRa's extremely

low bitrate, long airtime, and strict duty-cycle regulations pose serious challenges to classical routing protocols. The AODV protocol, originally created for higher-bandwidth MANETs, generates frequent RREQ/RREP control traffic that would saturate LoRa channels and lead to broadcast congestion [2]. Routing overhead becomes a bottleneck, significantly degrading throughput and latency in multi-hop LoRa networks [3].

In this context, end-to-end performance is commonly characterized by packet delivery ratio (PDR) and multi-hop latency; accordingly, we also provide an analytical best-case LoS estimate of these metrics for a representative 3-hop path (in addition to our measured routing overhead reduction).

Security adds another layer of complexity. LoRaWAN specifies AES-128 encryption, but UART-based LoRa modules such as the RYLR890 do not provide built-in end-to-end payload confidentiality for standalone multi-hop operation. Therefore, infrastructure-less LoRa mesh systems require application-layer cryptography to ensure end-to-end payload confidentiality [4], [5].

Despite existing attempts to explore LoRa mesh routing, prior work has not demonstrated, on Arduino-class devices, an integrated prototype that combines (1) reduced-broadcast AODV-style route discovery and (2) end-to-end payload confidentiality using UART-based LoRa modules.

To address this gap, we introduce **RelayCom**, a standalone LoRa mesh communication system designed for autonomous and off-grid environments with end-to-end payload confidentiality. RelayCom makes two main contributions:

- **A LoRa-aware adaptation of AODV**, replacing classical broadcast-based RREQ initiation with a novel unicast-first RREQ strategy. The node attempts a direct RREQ to the last-known next hop in its routing table before resorting to broadcast, reducing broadcast transmissions and minimizing LoRa channel congestion.
- **A fully embedded AES-CTR encryption engine on Arduino Nano**. This lightweight architecture provides end-to-end payload confidentiality without relying on

LoRaWAN servers, while remaining feasible on sub-16 MHz microcontrollers.

These contributions are implemented in a complete working prototype that integrates firmware modules (routing manager, LoRa handler, state machine), a user interface, and real hardware nodes. Experimental results obtained on a three-node testbed validate the feasibility of encrypted multi-hop routing and route optimization under realistic indoor conditions.

This paper is organized as follows: Section II reviews related work, Section III describes the RelayCom design and implementation, Section IV presents the experimental results, Section V discusses limitations and security considerations, and Section VI concludes with future work.

II. RELATED WORK

LoRa has attracted significant research interest for long-range, low-power communication, particularly in scenarios where network infrastructure is absent or unreliable. Early efforts focused primarily on LoRaWAN, where communication follows a star topology through centralized gateways [1]. While suitable for IoT sensing, LoRaWAN cannot support multi-hop routing, which limits its applicability in infrastructure-less deployments.

To overcome this limitation, several studies have explored LoRa mesh networking. Durand demonstrated the feasibility of multi-hop LoRa routing in indoor environments [2]. Despite promising results, these systems rely on comparatively powerful hardware and do not address the resource constraints typical of Arduino-class microcontrollers.

More recently, routing approaches over LoRa have been evaluated, including AODV-based designs. Li et al. showed that traditional flooding-based RREQ broadcasts increase channel occupancy, collisions, and energy waste in multi-hop LoRa settings [6]. However, prior studies do not provide a fully embedded implementation on ultra-constrained nodes with LoRa-aware optimizations that reduce broadcast reliance.

Security in LoRa systems has also been widely examined. While LoRaWAN includes AES-128 encryption at the MAC layer, UART-based LoRa modules such as the RYLR890 lack integrated end-to-end security mechanisms for standalone deployments, and therefore do not guarantee payload confidentiality [7]. Consequently, application-layer encryption is typically required for end-to-end payload confidentiality in standalone deployments [8]. RelayCom adopts AES-CTR encryption directly on MCU-class devices, consistent with lightweight embedded implementations for constrained IoT nodes [9].

In addition to integrating routing and confidentiality on constrained hardware, RelayCom evaluates a broadcast-first AODV baseline to quantify the reduction in route-discovery broadcast transmissions.

III. METHODS AND SYSTEM DESIGN

A. Overview

RelayCom is designed as a fully autonomous LoRa mesh communication system operating on ultra-constrained hard-

ware and optimized for off-grid environments. The methodology integrates three core components: (1) a minimal-cost embedded hardware platform combining an Arduino Nano and a Reyax RYLR890 LoRa transceiver; (2) a LoRa-adapted AODV routing protocol featuring a novel direct-unicast-first RREQ strategy; and (3) an end-to-end AES-CTR encryption pipeline executed directly on the microcontroller. This design aligns with recent efforts on low-cost LoRa mesh networking [10] and lightweight IoT cryptography, while addressing their limitations under strict energy, memory, and bandwidth constraints.

B. Hardware Architecture

Each RelayCom node is built using an Arduino Nano microcontroller, a Reyax RYLR890 LoRa module [11], a 4×4 keypad, and an SH1106 OLED display. Figure 1 illustrates the core hardware components. Similar embedded LoRa testbeds have been used in recent studies evaluating resource-constrained mesh architectures [12].

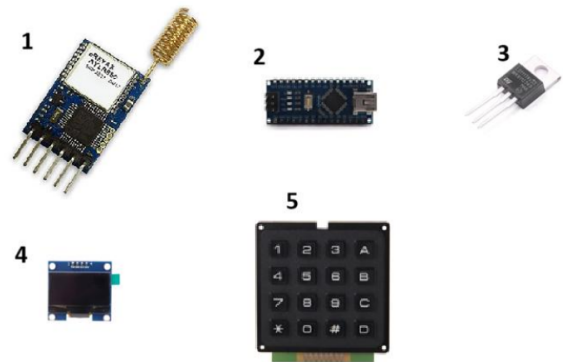


Fig. 1. Key components of a RelayCom node.

Internally, the Arduino communicates with the LoRa module over UART, powers the RYLR890 through a 3.3 V regulator, and interfaces with the OLED via I²C. This approach follows established low-power embedded LoRa design practices. The complete circuit schematic is shown in Figure 2.

This hardware design maintains low complexity while enabling multi-hop routing, end-to-end payload encryption (confidentiality), and real-time user interaction through the integrated keypad and OLED interface.

C. Firmware Architecture

RelayCom's firmware adopts a modular, event-driven structure optimized for deterministic operation on microcontrollers with limited RAM and Flash. Figure 3 presents the high-level firmware architecture.

The architecture consists of five coordinated modules:

- **LoRaHandler:** abstracts AT-command communication with the RYLR890 and parses +RCV frames.
- **RoutingManager:** implements the LoRa-adapted AODV protocol, including the direct-unicast RREQ mechanism, duplicate suppression, and route lifetime management.

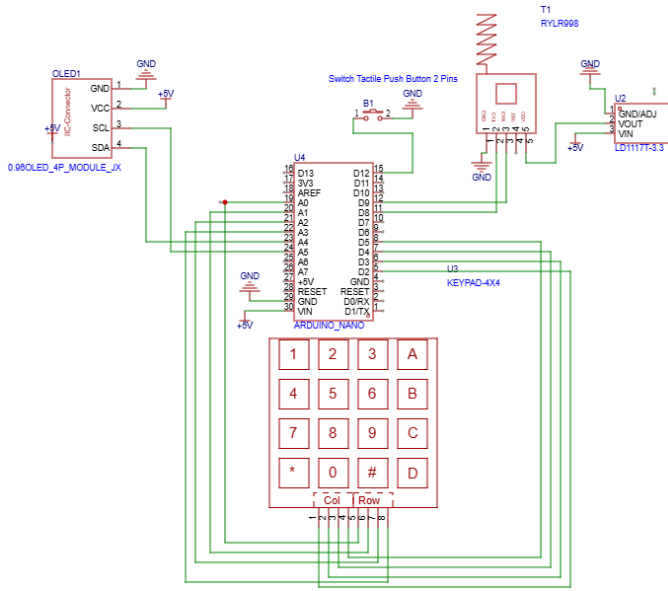


Fig. 2. Circuit schematic of a RelayCom node.

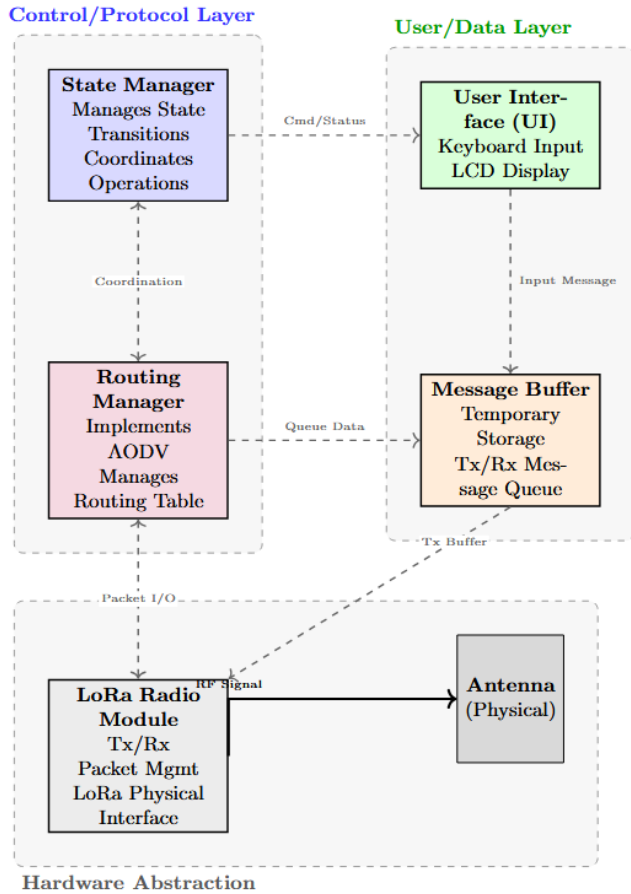


Fig. 3. Software-Hardware Architecture of the LoRa Mesh Node.

- **StateManager:** controls the node through a finite-state machine, ensuring clean transitions between route discovery, transmission, and error recovery.

- **SecurityModule:** performs AES-CTR encryption/decryption using a pre-shared key and a per-message counter-based nonce to ensure nonce uniqueness across transmissions, in line with lightweight AES implementations for constrained IoT devices [9]. AES-CTR provides confidentiality only; integrity, authentication, and replay protection are not provided in the current prototype.
- **UI Module:** handles keypad input and OLED updates through non-blocking routines.

D. LoRa Communication Layer

The LoRa communication layer handles all interactions with the RYLR890 transceiver [7]. UART communication is configured at 115200 bps, and nodes initialize radio settings (frequency, spreading factor, bandwidth, coding rate, and TX power) through AT commands.

TABLE I
LoRa PHY CONFIGURATION AND EXPERIMENTAL SETUP (TESTBED)

Parameter	Value
LoRa module	Reyax RYLR890
MCU	Arduino Nano (ATmega328P, 16 MHz)
UART baud rate	115200 bps
Frequency / region	868 MHz
Spreading factor (SF)	SF12 (default)
Bandwidth (BW)	125 kHz (default)
Coding rate (CR)	4/5 (default)
TX power	15 dBm (default)
Payload size (bytes)	8–32 bytes (user text + routing metadata)
Topology	3 nodes, linear (1–2–3); Node 1 cannot reach Node 3 directly
Trials	10 trials per configuration; nodes reset before each trial
Environment	Indoor short-range and indoor→outdoor obstructed links

Incoming LoRa frames are parsed into structured fields (source address, RSSI, SNR, payload) and forwarded to the routing layer. Outgoing packets (RREQ, RREP, MSG) are formatted as application-layer payloads and transmitted through the transceiver in transparent mode.

1) *Analytical PDR/Latency Estimation Model (Best-Case LoS):* To complement the measured routing-overhead results, we use a simple analytical model to estimate best-case end-to-end PDR and latency for a multi-hop path. We model end-to-end PDR over H hops as $PDR_{e2e} = p^H$, where p is the assumed independent per-hop delivery probability under clean LoS conditions. End-to-end latency is approximated by store-and-forward relaying, $L_{e2e} \approx H \cdot (ToA + T_{fwd})$, where ToA is the LoRa time-on-air under the PHY configuration in Table I, and T_{fwd} is a small per-hop forwarding/processing delay. These estimates are provided as a sanity-check reference and are not experimentally validated in this study.

E. LoRa-Adapted AODV Routing Protocol

RelayCom implements a lightweight AODV variant tailored for the low throughput and long airtime of LoRa. Figure 4 summarizes the RREQ-RREP route discovery process.

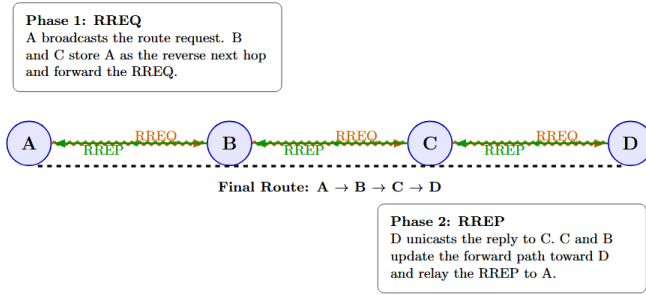


Fig. 4. AODV route establishment: RREQ travels from A to D (orange), and the RREP returns from D to A (green), forming the A→B→C→D route.

Novel Adaptation: Direct-Unicast-First RREQ Instead of immediately broadcasting RREQs (as in classical AODV), RelayCom first attempts a unicast RREQ to the last-known next hop for the destination in its routing table (if a valid entry exists). This approach is inspired by LoRa-aware optimizations proposed in [13] but extends them by integrating timeout-driven escalation logic. Only if the unicast attempt fails does the node escalate to a broadcast RREQ. This reduces broadcast transmissions and collision probability, which is critical under LoRa's low bitrate and duty-cycle constraints.

RREQs and RREPs include hop counts, sequence numbers, and lifetime fields. Forward and reverse routes are installed according to AODV principles, with timers ensuring route freshness. These lightweight routing behaviors follow best practices in recent embedded AODV implementations [14].

IV. RESULTS

This section evaluates RelayCom on (i) route discovery feasibility in direct and multi-hop scenarios, (ii) reduction in route-discovery broadcast transmissions using the proposed direct-unicast-first AODV adaptation, and (iii) verification of end-to-end encrypted message delivery over one and two hops.

A. Route Discovery Performance

RelayCom successfully established routes in both direct and multi-hop scenarios. In the direct case (Node 2 → Node 3), route discovery was completed reliably with correct neighbor identification. In the multi-hop scenario (Node 1 → Node 2 → Node 3), routes were successfully discovered across multiple hops, with hop count accurately reported and routing entries correctly installed at all participating nodes.

We further verified end-to-end encrypted message forwarding over one and two hops by exchanging application payloads that were encrypted at the sender and correctly decrypted at the destination.

To evaluate the impact of environmental conditions on LoRa signal propagation, RSSI and SNR measurements were collected for two representative scenarios: a short-range indoor

link and an indoor-to-outdoor link with significant physical obstructions. These scenarios reflect realistic deployment conditions for low-power LoRa mesh networks, where nodes may operate inside buildings while communicating with mobile or outdoor devices.

1) *Indoor (short-range)*: In this scenario, both nodes were placed indoors at a separation of approximately 1.5 m. The environment had no major obstructions, allowing for an almost ideal short-range LoRa link. As expected, the RSSI and SNR values were strong, indicating a clean, high-quality channel typical of close-proximity indoor communication.

2) *Indoor → Outdoor (walls + vehicle)*: In this configuration, the transmitting node was located inside a residential apartment, positioned 1–2 m from a window. The receiving node was placed outdoors inside a parked vehicle. The radio path therefore included multiple attenuation sources: one structural wall, window glass, and the metal chassis of the vehicle, in addition to an increased distance of approximately 350 m. These factors created a partially obstructed, non-line-of-sight environment representative of real-world usage. Despite these conditions, the received RSSI and SNR values remained within the decodable range of LoRa modulation, demonstrating the robustness of the communication link. Table II summarizes the measurements across both scenarios. RSSI and SNR values are reported as provided by the RYLR890 module in the +RCV metadata.

TABLE II
RSSI AND SNR VARIATION ACROSS ENVIRONMENT TYPES

Environment	Distance	RSSI (dBm)	SNR (dB)
Indoor (short-range)	~1.5 m	−43	35
Indoor → Outdoor (walls + vehicle)	~350 m	−100 to −115	−3 to 3

B. Multi-Hop Routing Table Verification

To validate correct multi-hop route installation, a three-node linear topology (1–2–3) was configured such that Node 1 could not directly reach Node 3. Route discovery was initiated at Node 1. Node 2 received the RREQ and forwarded it to Node 3, which generated the RREP and returned it to Node 1 through Node 2.

After route establishment, each node displayed its routing table on the OLED interface. The results confirm that the forward and reverse paths were correctly installed. Node 1 selected Node 2 as the next hop toward Node 3 with a hop count of two. Node 2 maintained direct entries for both neighbors, while Node 3 installed the reverse path to Node 1 through Node 2. Fig. 5 shows the routing tables as displayed on the OLED screens of the three nodes.

C. Reduction in Broadcast Overhead

For comparison, we implemented a baseline corresponding to *classical AODV* route discovery on the same platform, where RREQ initiation is *broadcast-first* (i.e., no initial unicast attempt). Both RelayCom and the baseline use identical LoRa



Fig. 5. Routing tables displayed on the OLED screens after successful multi-hop route discovery in the 1–2–3 topology. Left: Node 1; Center: Node 2; Right: Node 3.

PHY settings (Table I), packet formats, and timeout/retry parameters to ensure a fair comparison. A key advantage of the proposed routing strategy is the reduction in RREQ broadcast transmissions. Across 10 repeated trials, classical AODV required 10 RREQ broadcast transmissions during the initial route-discovery phase, whereas RelayCom required only 3 due to the unicast-first escalation logic. This corresponds to a 70% reduction in RREQ broadcast transmissions, which can reduce channel occupancy and collision probability—an essential consideration given the long airtime of LoRa frames [15]. We report the number of RREQ *broadcast* transmissions issued by the source during route discovery (excluding unicast attempts).

D. Estimated End-to-End PDR and Latency (Best-Case LoS)

To provide a sanity-check reference for multi-hop performance under ideal conditions, we report an analytical best-case estimate of end-to-end packet delivery ratio (PDR) and latency for a 3-hop path with approximately 500 m per hop in line-of-sight (LoS) conditions. These values are illustrative estimates and are not experimentally validated in this study.

Estimated PDR: Assuming an independent per-hop delivery probability $p = 0.99$ in clean LoS conditions, the end-to-end delivery probability over $H = 3$ hops is:

$$\text{PDR}_{e2e} = p^H = 0.99^3 \approx 0.970 \text{ (97.0\%)}$$

Estimated latency: End-to-end latency is approximated by store-and-forward relaying:

$$L_{e2e} \approx H \cdot (\text{ToA} + T_{fwd}),$$

where ToA is the LoRa time-on-air per packet under the PHY configuration in Table I, and T_{fwd} is a small per-hop forwarding/processing overhead. The ToA value is computed using the standard LoRa time-on-air formulation (explicit header mode, CRC enabled). For a representative payload in our range (e.g., 16 bytes), SF12/BW125k/CR4/5 yields $\text{ToA} \approx 1.32$ s, and using $T_{fwd} \approx 50$ ms gives:

$$L_{e2e} \approx 3 \cdot (1.32 + 0.050) \text{ s} \approx 4.11 \text{ s}.$$

Figs. 6 and 7 illustrate the direct and multi-hop route discovery behaviors, respectively.

V. DISCUSSION

The experimental evaluation confirms that the proposed LoRa-adapted AODV protocol and lightweight AES-CTR confidentiality layer can operate on ultra-constrained hardware. The direct-unicast-first RREQ mechanism reduced broadcast

TABLE III
ANALYTICAL BEST-CASE LOS ESTIMATE FOR A 3-HOP PATH (NOT EXPERIMENTALLY VALIDATED).

Parameter	Value
Number of hops (H)	3
Distance per hop	~500 m (LoS)
Assumed per-hop PDR (p)	0.99
Estimated end-to-end PDR (p^H)	0.970
LoRa PHY (from Table I)	SF12 / BW125k / CR4/5
Representative payload	16 bytes
Time-on-air (ToA)	1.32 s
Forwarding delay (T_{fwd})	50 ms
Estimated end-to-end latency (L_{e2e})	4.11 s



Fig. 6. Direct route discovery when Node 3 is within the communication range of Node 1. The unicast-first RREQ mechanism establishes a single-hop route without intermediate forwarding. Top-left: Destination selection; Top-right: Unicast RREQ transmission; Bottom-left: Direct route found (1 hop); Bottom-right: Installed routing table.



Fig. 7. Multi-hop route discovery when Node 3 is outside the direct communication range of Node 1. RREQ forwarding via Node 2 results in a two-hop route with correct routing-table installation. Top-left: Broadcast RREQ initiated; Top-right: Multi-hop route found (2 hops); Bottom-left: Route confirmation; Bottom-right: Installed routing table.

transmissions by approximately 70%, demonstrating a clear advantage over classical AODV behavior in low-throughput networks. This aligns with observations from recent studies emphasizing LoRa's susceptibility to congestion and collisions under broadcast-heavy routing strategies [12].

The system's ability to reliably establish multi-hop routes

and deliver encrypted messages reflects the practicality of integrating application-layer encryption in a standalone LoRa mesh. This layered approach complements previous security analyses on LoRa end devices [8], [16], while demonstrating that payload confidentiality can be achieved without LoRaWAN infrastructure. In contrast to works that offload cryptographic operations to more capable devices [17], RelayCom performs AES-CTR directly on an MCU-class platform while maintaining responsive routing behavior in our testbed.

These PDR/latency values represent best-case LoS behavior; in practical deployments, interference, duty-cycle constraints, retransmissions, and obstructions can substantially reduce PDR and increase latency.

The prototype's behavior under indoor obstruction confirms that RSSI/SNR levels remain sufficient for multi-hop routing. Nonetheless, the system inherits inherent limitations from the underlying hardware: absence of MAC-layer collision avoidance, limited route-table size, and the potential for replay attacks due to static keying. In the current prototype, the AES key is configured statically at both sender and receiver for controlled testing and repeatability. We emphasize that AES-CTR provides confidentiality only; integrity, sender authentication, and replay protection are not guaranteed in the current prototype. Future iterations will adopt authenticated encryption (or CTR+MAC) with counter-based nonces and explore duty-cycle-aware backoff strategies and secure-element-based key storage, consistent with emerging best practices in LoRa mesh deployments [2], [6].

Overall, RelayCom demonstrates that an infrastructure-free LoRa mesh with end-to-end payload confidentiality is feasible using only commodity microcontrollers, although further optimization is required for large-scale deployments.

The presented results demonstrate feasibility on a small-scale testbed rather than scalability. As the number of nodes increases, contention, routing overhead, and latency are expected to grow due to LoRa's limited bandwidth and duty-cycle constraints. While the analytical PDR and latency estimates provide useful baseline insights under ideal LoS conditions, they cannot fully capture real network dynamics. A comprehensive experimental evaluation of end-to-end delay and packet delivery ratio under larger-scale deployments remains part of future work. Furthermore, in practical deployments, the proposed system is particularly suited for infrastructure-less scenarios such as emergency response, rural connectivity, and temporary field networks. In such environments, nodes can be rapidly deployed to form a self-organizing communication system without relying on centralized infrastructure, enabling secure and resilient message exchange under constrained conditions. Although this work focuses primarily on validating feasibility and routing efficiency through broadcast reduction, additional performance metrics such as end-to-end latency and packet delivery ratio were not experimentally evaluated in this prototype. This choice was motivated by the constraints of the initial testbed and the objective of demonstrating a functional, lightweight implementation on MCU-class devices. Future work will include systematic measurement of these metrics

under varying network sizes, traffic loads, and environmental conditions to provide a more comprehensive performance evaluation.

VI. CONCLUSION

This work presented RelayCom, a fully standalone LoRa mesh communication system combining (i) a lightweight hardware platform, (ii) an AODV-derived routing protocol tailored for LoRa's low data rate, and (iii) an end-to-end AES-CTR confidentiality mechanism implemented on microcontrollers. The proposed direct-unicast-first RREQ strategy significantly reduced route-discovery broadcast transmissions, improving route discovery efficiency while maintaining correct multi-hop operation in the evaluated scenarios.

Experimental results demonstrated consistent encrypted message delivery and stable link-quality indicators, confirming the feasibility of off-grid LoRa mesh networking with end-to-end payload confidentiality using low-cost embedded nodes. These contributions provide a practical foundation for resilient communications in remote, emergency, or infrastructure-degraded areas.

In the current design, route selection is primarily based on hop count, following classical AODV semantics. Future work will investigate multi-factor routing metrics that incorporate link-quality indicators (RSSI/SNR), local noise conditions, and node-level parameters such as temperature or energy availability. Such a composite metric could enable more robust path selection and improve load distribution across the mesh. We also plan to explore additional authenticated encryption hardening and antenna optimization to further strengthen range, robustness, and resistance to hostile or noisy environments.

Future work will additionally address scalable key management, authenticated encryption, adaptive duty-cycle control, and validation on larger network topologies to extend RelayCom toward production-grade deployments.

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